Design and Implementation of Color Two-Dimension Barcode with high compression ratio for Chinese characters

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Abstract—**Concerning the prevailing two-dimensional barcode with insufficient capacity and low compression ratio for Chinese character, a new color two-dimensional barcode is proposed here with large capacity and high compression ratio for Chinese character. Referring to Data Matrix barcode standard, the coding rule of Chinese character is considered in this paper. Through redesigning data coding method and decoding method, a color two-dimension barcode with both high compression ratio for Chinese character and relatively high capacity to store information is described. Compared with traditional black-and-white two-dimension barcode, this color two-dimension barcode can achieve better performance in compression ratio.**

Keywords- Color Two-Dimension Barcode; Chinese Character; Two-Dimension Barcode

I. INTRODUCTION

At present, the black-and-white two-dimensional bar code technology has developed more mature. American National Standards Institute (ANSI) developed the international standards of two-dimensional bar code, including Data Matrix, PDF417, QR codes[1][3]. However, with the increase of information, as well as expanding the field of bar code applications, bar code need to be stored more information. The current black-and-white two-dimensional bar code can only use two colors to describe the information and this limited the increase of the information capacity of bar code in a larger extent. Under this background, the color bar code opens a new research field. Microsoft Research Institute proposed a full-color two-dimensional bar code. There are two kinds of versions with four-color barcode and eight-color barcode. The smallest recognition unit is a triangle. As opposed to black-and-white two-dimensional bar code, this color bar code can raise approximately 2-3 times data density[2]. The company of ColorZip in South Korean is also promoting the value-added business based on color bar. The current prevalent two-dimensional bar codes, including Microsoft Research Institute, are mainly developed for English letter and digital number. For the Chinese characters coding, the Chinese character internal code are used. This coding method has low efficiency because all Chinese characters are coded in the same coding length. So in this paper, referring to the Data Matrix standards, a color two-dimensional bar code with high compression ratio for Chinese characters is designed and achieved.

In this paper, the design principle of color two-dimensional bar code is introduced. Then the coding principle is described. The performance of this bar code is verified. Finally, the decoding principle is proposed and the test result is given.

II. PRINCIPLE OF COLOR TWO -DIMENSIONAL BAR CODE

A. *Information Capacity*

In the black-and-white two-dimensional bar encoded method, the data will be transformed into binary data stream. Then these binary data value, 0 and 1, are be presented with black-and-white symbols. For each module can present 0 or 1, each module can express two kinds different information. So in the black-and-white two-dimensional bar with n symbol modules, the information capacity will be 2^n. In the color bar code, supposing if the color amount is k and different color corresponds to 0,1,2,……,k, the capacity of color bar code with n modules will reach k^n. So the color amount in the color bar code is large than 2, it will increase the information capacity with certain encoding rules. In order to make it easy to identify, the strong contrast colors are selected. The six kinds color of black, white, red, green, yellow, blue are selected to express data information in this paper.

B. *The structure of color bar code*

The structure of color bar code can be divided into two parts: function image and data image. The function image is used to position and correct the bar code. The data image is used to express the data information encoded. The color bar code for Chinese characters is composed of square modules arranged with rules. It is showed in figure 1 below.

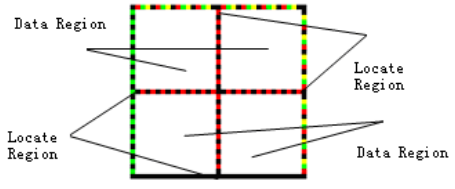


Figure 1. The structure of color bar code

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III. ENCODING PRINCIPLE IN COLOR BAR CODE

A. Encoding process

There are three steps in the encoding process of color bar code: encoding the data information, generating error-correcting code[4], and generating symbol process. Firstly, according to the data information encoding mode, the data is transformed into data stream. Meanwhile, the information of length of data stream is added into the head of data stream. Then each four bits in the data stream form a code and the corresponding error-correcting code is generated through Reed-Solomon algorithm. The error-correcting code is added to the end of the data stream. Finally, the all data codes will be transformed into symbols and the function images are added too. The color bar code is generated. The figure 2 shows the encoding process.

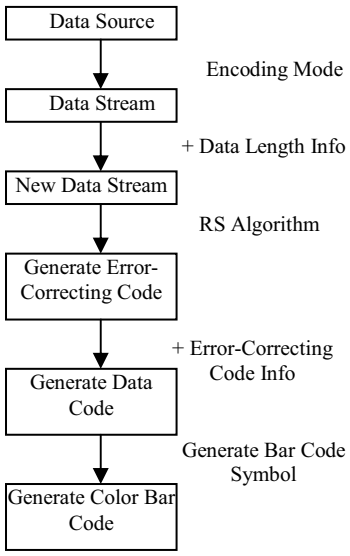


Figure 2. The encoding process

B. Encoding mode

- Chinese characters encoding mode

In Chinese character set GB, there are 6763 Chinese characters and 682 non-Chinese graphic symbols. But through statistical analyzing, only 3500 Chinese characters are commonly used. The most commonly used Chinese characters are 600 and they almost cover 80% of Chinese character set GB. If these Chinese characters are presented with the different colors, at least five different colors are used. That means the largest number in the five-base system is the 444445, and it is equal to 312410 in the decimal system. These 3124 Chinese characters almost cover the 99% of Chinese character set GB. In order to add an identification bit, the six-base system is adopted. That means the 6 different colors are used to express the data information.

According to the frequency table of Chinese characters from Tsinghua University, the Chinese character encoding rules are designed. It is showed in table 1 below.

The encoding rule of Chinese character's internal code is designed for those less commonly used Chinese characters and less commonly used punctuations.

TABLE I. CHINESE CHARACTER ENCODING RULES

Using Frequency	**Most ommonly used char 600**	**Commonly used punctuation 25**	**Commonly used char 2500**
Code Value	0～599	600～624	625～3124
Encoding Rules	Chinese Characters sequence number in frequency table of GB will be transformed into a number in five-base system. If the number is larger than 625 (That is 10000_5-1), the encoded number should be added identification code '5' at the front of the 5-places digit and added identification code '55' at the front of the 6-places digit. For example, if the encoded number is '11111", it should be added 5 at the front of it and it is "511111'. If the encoded number is '111111', it should be added 55 at the front of it and it is '55111111'.		

- Encoding rule of Chinese character's internal code

TABLE II. ENCODING RULE OF INTERNAL CODE

Type	Encoding rule	Range of Encoded digit
Chinese character's internal code	High place type in internal code *16+low place type in internal code —>a digit with 7 bits in five- base system	20776~65278

In this situation, the identification code is '50' and the whole encoded digit is 9 bits.

- Encoding mode for single ASCII

TABLE III. ENCODING RULE OF ASCII

Type	Encoding rule	Range of Encoded digit
Single ASCII	Single ASCII is transformed to five-base digit	0000~1002

In this situation, the identification code is '50' and the whole encoded digit is 6 bits.

- Encoding mode for successive digits

Table 4: Encoding rule of successive digits

Type	Encoding rule
Consecutive even number of digits	Each two digits form a five-base code with three bits.
Consecutive odd number of digits	Each two digits form a five-base code with three bits. The last digit forms a five-base code with two bits.

In this situation, the identification code is '515' at the beginning and at the end of this code, the identification code '5' is appended.

- Encoding rule for consecutive ASCII

The encoding rule for consecutive ASCII is divided into two sets: S and R. For example, the consecutive ASCII 'AB' is encoded, the encoded code will be 'ab'. The encoding rule is showed in table 5 and table 6.

TABLE IV. SET S AND SET R

Set	S	R
elements	Upper and lower case letters and commas, periods and spaces	The rest 37 elements that are not below to S
Code value	0～54	0～72

TABLE V. ENCODING RULES FOR CONSECUTIVE ASCII

Type	Encoding rule	Identification code
A∈S, B∈S	(a*55+b)—> five-base digit	None
A∈S, B∉ S ||B=∅	a—> five-base digit with 3 bits	None
A∈R	a—> five-base digit with 3 bits	Adding '5' at the front of the encoded digit

In this situation, the identification code '505' is added at the beginning and at the end of this code, the identification code '505' is appended.

C. *Generating symbol*

In order to generate bar code symbol, the encoded digit is mapped to these colors below in table 7:

TABLE VI. DIGIT- COLOR MAPPING TABLE

Digit	0	1	2	3	4	5
Color	black	white	blue	green	red	yellow

In the mapping process of the whole color image, the digits are transformed from left to right and from bottom to top. It is the same for the each data region. A mapping case of '3205' is showed in figure 3.

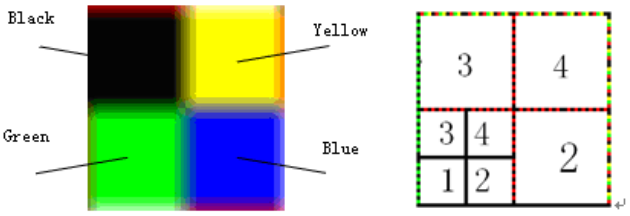


Figure 3. Process of generating symbol

D. *Performance comparison*

Comparing to the black-white bar code, the color bar code has a great improvement in the data capacity and data compression ratio.

The data capacity of the black-white bar code and the data capacity of color bar code are compared below. Supposing there are k colors and n modules in a color bar code, the data capacity is . But for the black-white bar code, the data capacity is . In this paper, there are 6 colors in the color bar code. So the data capacity is . The information capacity of two-dimensional bar code on the contrast with the black-and-white bicolor has greater advantage.

For comparing the data compression ration between the black-white bar code and color bar code, the news below can be used to test.

12月13日(北京时间9时00分左右)，全国第二届省级示范性软件学院院长联席会在我校举行。

In this news, there are total 36 Chinese characters and Chinese punctuations, and 8 ASCII. According to the encoding rules of color bar code, the encoded code will reach 194 bits. The detailed statistics of encoding modes is showed in table 8.

TABLE VII. THE STATISTICS OF ENCODING MODES IN THE CASE

Encoding mode	Chinese char with 4 bits	Chinese char with 6 bits	digital	Single ASCII
Characters numbers	33	3	2	6

If the news is encoded according to the Data Matrix code, the 36 Chinese characters and Chinese punctuations will be encoded. In this encoding mode, each character will be 16 bits. In addition, the length of the encoded code with 16 bits will be added. So the encoded Chinese characters will be:

$36\times16+16=592$ bits

For digital and ASCII characters, in the optimal coding, they at least need:

$2\times4+5.33\times6\approx40$ bits

So the total bits will be

$592+40=632$ bits

The compression ratio is $632\div194\approx3.25$. So the data compression ratio of color bar code is better than the that of black-white bar code.

The news is encoded separately with color bar code and black-white bar code (Data Matrix) and the encoded symbols are showed in the figure 4 below.



Figure 4. Comparison with color bar code and Data Matrix for the encoded news

IV. DECODING PRINCIPLE

The decoding principle is the inverse process of encoding. There are main three steps in the decoding process.

A. *Positioning of image*

The color bar code image can be positioned through the function image. In the color bar code, there are three basic colors: red, green and blue. In one of three color channel, for example, the red channel, the function image can be used to positioning through Hough algorithm. When the positioning

under the red channel can not be achieved, the function images under green channel and blue channel will be used. It improves the positioning method in Data Matrix code.

B. Data stream transformed from color code

In this step, the bar code size will be detected and the data stream will be transformed from color code through transformation algorithm. The bar code size can be calculated by counting the numbers of changes in a row or a column of location coordinates. Here the bar code size is supposed as n. Then using the vertex coordinates getting from the positioning process, the upper left corner vertex coordinate $P_1(x_1, y_1)$, the upper right corner vertex coordinate $P_2(x_2, y_1)$, the low left corner vertex coordinate $P_3(x_1, y_2)$, the low right corner vertex coordinate $P_4(x_2, y_2)$, the each module size can be calculated:

$$eachBlockSize = (x_2 - x_1)/n$$

or

$$eachBlockSize = (y_2 - y_1)/n$$

For the module located at row i and column j, the border coordinates are:

The X-axis direction:

$XposBegin = x_1 + i*eachBlockSize$,

$XposEnd = x_1 + (i+1)*eachBlockSize$

The Y-axis direction:

$YposBegin = y_1 + j*eachBlockSize$,

$YposEnd = y_1 + (j+1)*eachBlockSize$

In order to avoid the color cast, some statistical methods are used here. Counting the number of pixels inside the module and selecting the color value which the most of pixels have as color value of that module. Then the color value is transformed into digital number. This method is more advanced than the way of selecting the pixel color in the middle of module as the module color.

In the symbolizing process, the data information is stored from the left to right, and from bottom to top. So in the inverse decoding process, the encoded data will be read in the same way and form the data stream.

C. Decoding data information

After the encoded code is verified by the error-correction code, the encoded data can be read. Firstly, the 5 digits at the beginning of the code will be read and then these digits are transformed into decimal number. Then the length of data will be decided. According to the different identifications, the different encoding rules are applied. The table 9 shows the different encoding rules and corresponding identification codes.

TABLE VIII. DATA ENCODING RULES AND IDENTIFICATIONS

Encoding mode	Identification code
Chinese char with 4 bits	The highest place is not 5
Single ASCII char	The highest place is 500，501
Digital number	The highest place is 515
Consecutive ASCII char	The highest place is 505
Chinese char with 5 bits	The highest place is 5 and it is not other modes.
Chinese char with 6 bits	The highest place is 55
Chinese char with 7 bits	The highest place is 502,503,504

V. CONCLUSIONS

Here two cases of encoding and decoding are illustrated. One is the website of Zhejiang University of Technology, http://www.zjut. edu.cn:80. Other is the Chinese name of Zhejiang University of Technology, 浙江工业大学. The website encoded and decode is showed in figure 5. The Chinese name of the University encoded and decoded is showed in figure 6.

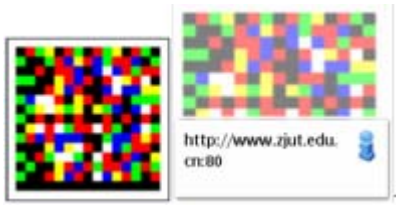


Figure 5. The encoded and decoded image of case 1

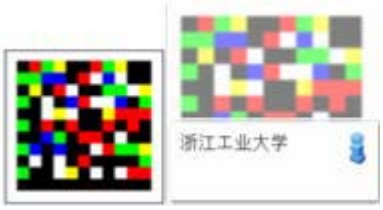


Figure 6. The encoded and decoded image of case 2

Nowadays, the decoding process is designed for the standard computer-generated images. In the actual image shot by mobile camera, the image may exist in the deformation and color cast. Meanwhile the positioning efficiency needs to be improved. At the same time, the speed of encoding and decoding is also required to be improved in further work in order to adapt to practical application.

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